INCH-POUND

MIL-DTL-24175/5C(SH)
14 July 2026
SUPERSEDING
MIL-P-24175/5B(SHIPS)
27 August 1971

DETAIL SPECIFICATION SHEET

PANEL, POWER SUPPLY AND CONTROL FOR SUBMARINE NAVIGATION LIGHTS, 10 AMPERES, 120 VOLTS AC, 60 HERTZ TOTALLY ENCLOSED, SYMBOL NO. 2550.1

This specification is approved for use by Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24175.

Part or identifying number (PIN)
M24175/5-001

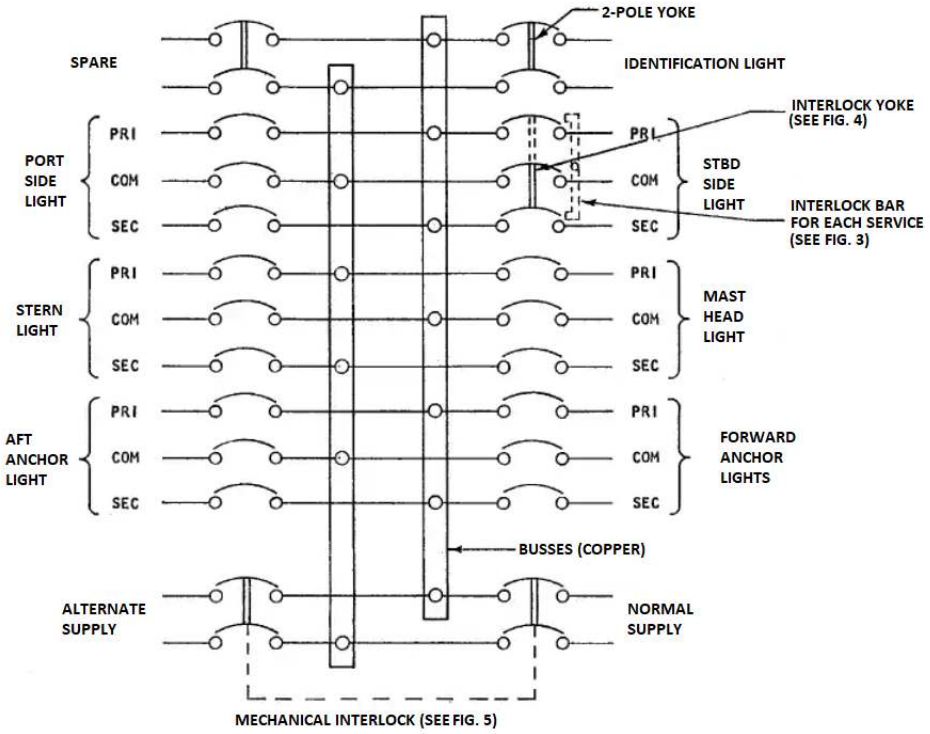


FIGURE 1. Panel, power supply and control for submarine navigation lights, totally enclosed, symbol no. 2550.1.

AMSC N/A FSC 6110

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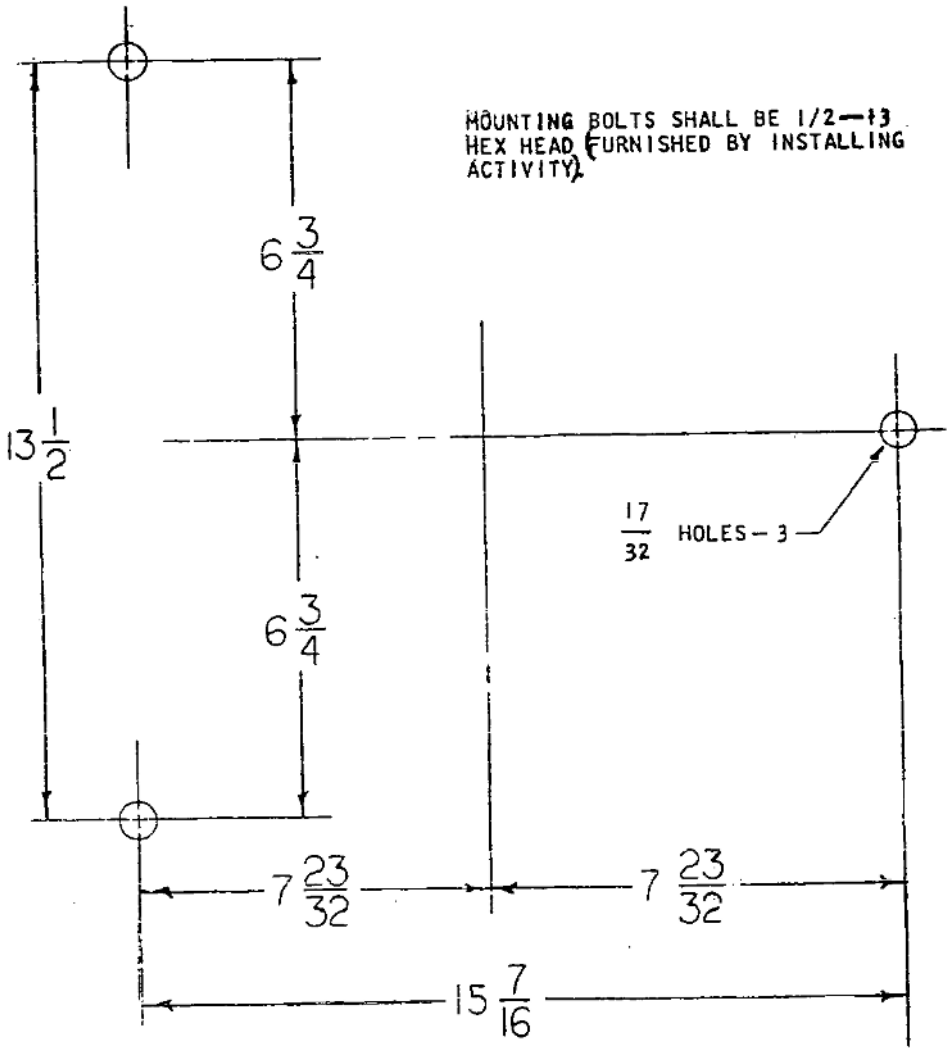


FIGURE 2. Mounting dimensions (inches).

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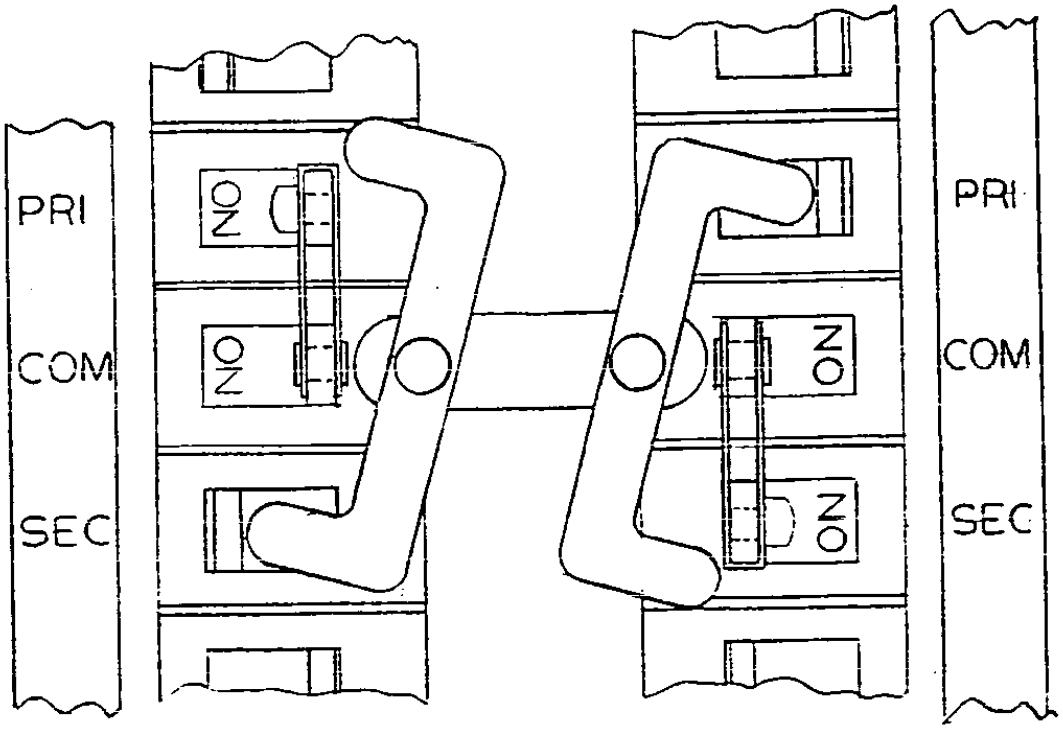


FIGURE 3. Interlocking bar.

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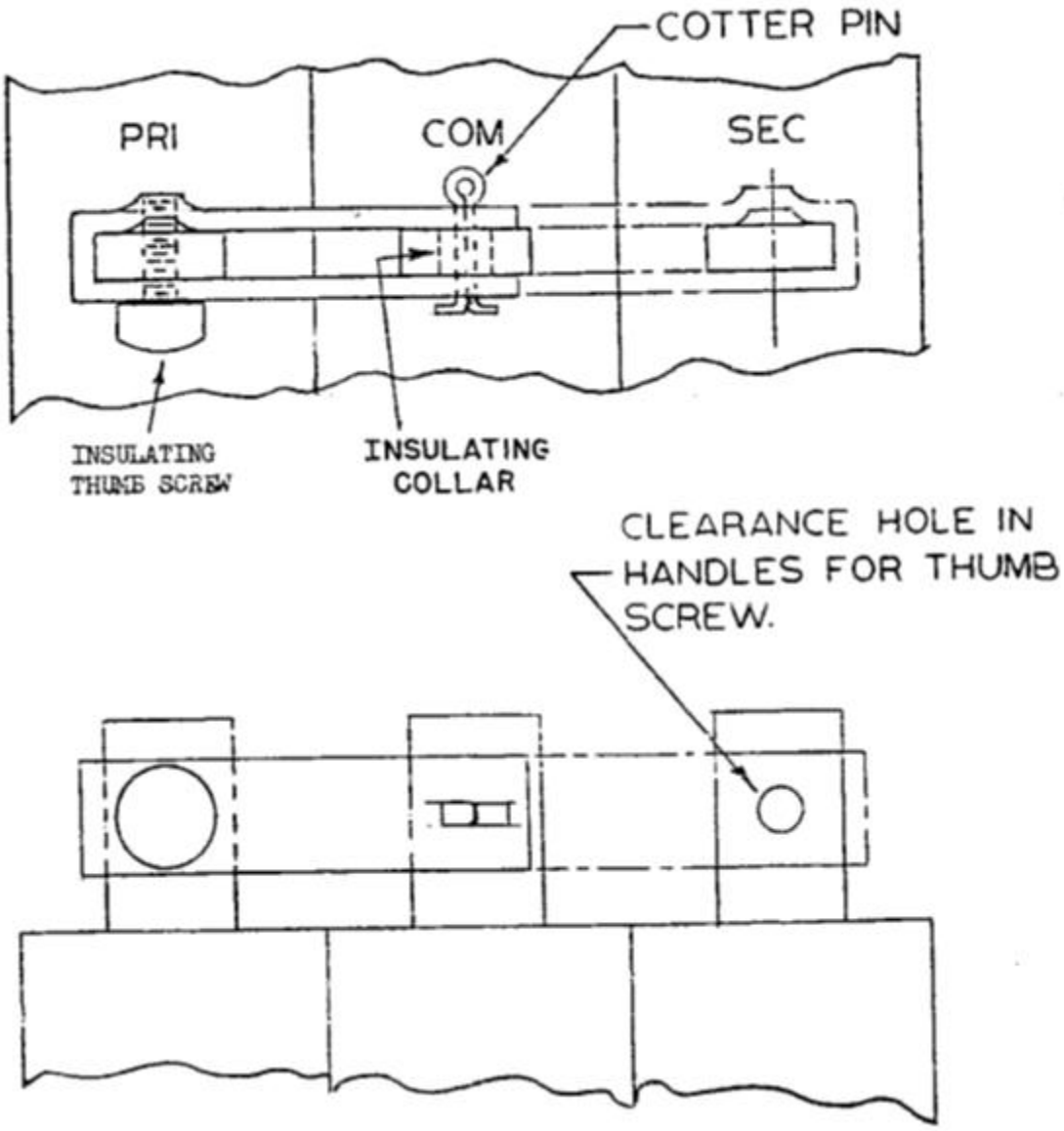


FIGURE 4. Interlocking handle yoke.

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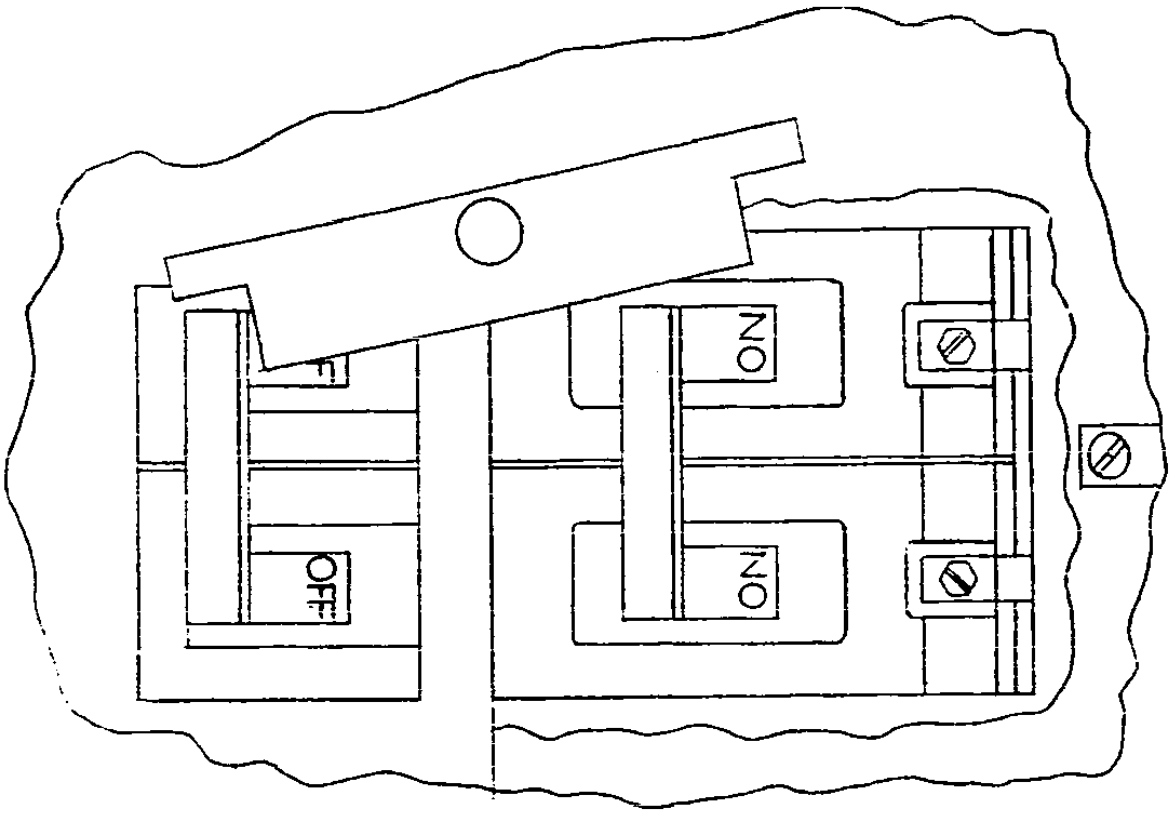


FIGURE 5. Mechanical interlocking bar.

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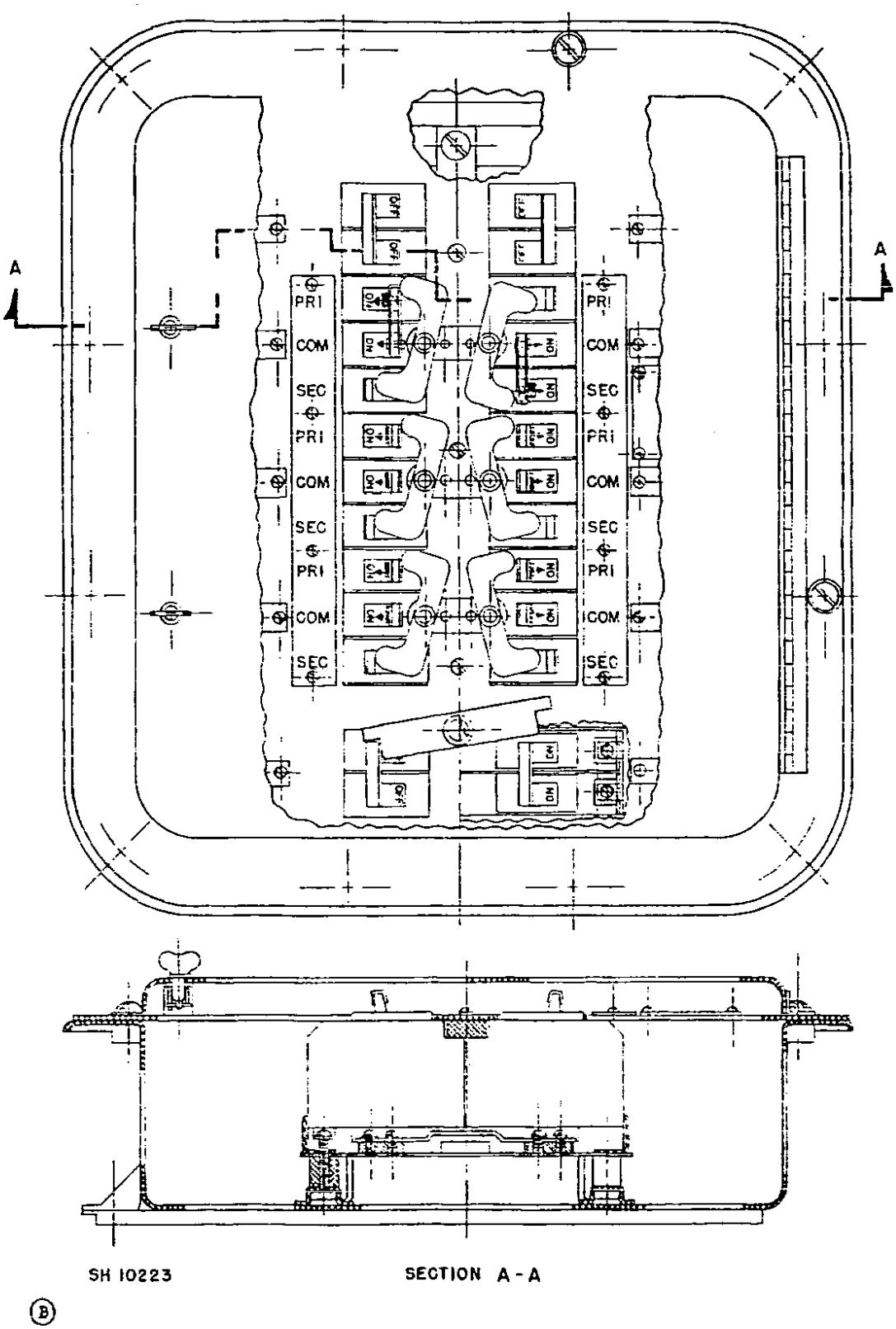


FIGURE 6. Typical enclosure arrangement.

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REQUIREMENTS:

Enclosure:	Overall dimensions shall be approximately 19½ inches high by 17½ inches wide by 5 inches deep, including cover, non-watertight construction. A hinged cover, for access to circuit breakers, shall be held shut by two quick-lock-unlock-type fasteners. Identification plates identifying function of groups of circuit breakers and individual breakers shall be mounted on breaker hold-down plate adjacent to respective breaker as shown on figures 1, 3, and 6.
Circuit breakers and busses:	Circuit breakers and busses shall be arranged and installed in the relative locations shown on figure 1. "ON" position of breakers shall be towards the busses in the center of the enclosure.
Circuit breakers, power supply:	Power supply circuit breakers shall be type NLB-1 of MIL-C-17588 (four required) and furnished by installing activity.
Circuit breakers, branch circuits:	Branch circuits circuit breakers shall be type ALB-1 with 5 ampere trip element of MIL-C-17588 (22 required) and furnished by installing activity.
Two-pole handle yokes:	Two-pole handle yokes shall be provided (see figure 4) for attachment to the handle of each circuit breaker marked "COM". Yoke and thumb screw shall permit either "PRI" and "COM" or "SEC" and "COM" circuit breakers to be locked together. This interlock is intended to prevent circuit breaker marked "COM" from being inadvertently left on when it is desired to completely de-energize the running light circuit (two required). Material for insulating thumb screw and insulation collar shall be plastic type MAI-60 of ASTM D5948 or a Group 01 nylon of ASTM D4066.
Interlocking handle yoke:	Interlocking handle yoke shall be provided (see figure 4) for attachment to the handle of each circuit breaker marked "COM". Yoke and thumb screw shall permit either "PRI" and "COM" or "SEC" and "COM" circuit breakers to be locked together. This interlock is intended to prevent circuit breaker marked "COM" from being inadvertently left on when it is desired to completely de-energize the running light circuit (six required). Material for insulating thumb screw and insulation collar shall be plastic type MAI-60 of ASTM D5948 or a Group 01 nylon of ASTM D4066.
Interlocking bar, for each service:	Interlocking bar for each service shall be installed (see figure 3) to prevent both "PRI" and "SEC" circuit breakers from being turned on at the same time (six required).
Mechanical interlocking bar:	Mechanical interlocking bar shall be installed between power supply circuit breakers (see figure 5) to prevent both alternate and normal power supplies from being turned on at the same time.
Wiring:	Wiring shall be as shown on figure 1. Only the wiring between the bus bars and the circuit breakers is required to be furnished by the manufacturer. Output wiring from circuit breakers is part of the vessel's electrical distribution system and not covered by this specification.
First article sample:	The first article sample when specified (see MIL-DTL-24175) shall be furnished complete as illustrated by figure 6 including mounting base, buswork lugs (when required), terminal assemblies, circuit breakers, interlocks, and so forth.

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CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Preparing activity:
Navy – SH
(Project 6110-2025-015)

Review activity:
DLA – GS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST at https://assist.dla.mil.

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